

Cyber Security Internship Report

Week 1: Security Assessment and Vulnerability Identification

In Week 1, the objective was to understand web application structure and identify basic security flaws. A Node.js-based mock application (Juice Shop or similar) was installed and executed locally.

The application included features such as user registration, login, and profile management. The system was accessed via `http://localhost:3000`.

Activities performed:

- Installation using npm and dependency setup
- Manual testing of input fields
- Observation of error handling behavior
- Identification of weak input validation

Key findings:

- Lack of strict input validation
- Passwords initially stored in insecure format (before fixes)
- No authentication tokens implemented

Conclusion:

This phase helped in understanding common web vulnerabilities and the importance of secure coding practices.

Week 2: Implementation of Security Measures

Week 2 focused on strengthening application security using industry-standard tools and techniques.

Implemented Security Features:

1. Input Validation:

Validator library was used to ensure correct email formats and sanitize user input.

2. Password Hashing:

bcrypt was implemented to hash passwords before storing them in the database. This ensures that even if data is leaked, passwords remain secure.

3. Authentication:

JWT (JSON Web Token) was used to implement secure login sessions. Tokens are generated after successful login and used to access protected routes.

4. HTTP Security:

Helmet.js was used to secure HTTP headers and protect against attacks such as XSS and clickjacking.

Conclusion:

This week significantly improved application security by implementing multiple layers of protection.

Week 3: Advanced Security and Final Reporting

Week 3 focused on advanced testing, monitoring, and documentation.

Penetration Testing:

Basic penetration testing was performed using manual input testing and Nmap scanning to detect open ports and running services.

Logging System:

Winston library was implemented to log application activities such as login attempts, errors, and system startup events. Logs are stored in a file for auditing purposes.

Security Checklist:

- Input validation enforced
- Passwords hashed and salted
- JWT authentication implemented
- HTTPS recommended for deployment
- Logging enabled for monitoring

Conclusion:

This phase completed the security lifecycle with monitoring, testing, and final documentation of the system.